RAJINDER SINGH

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SUMMARY

B.E. Mechanical Engineering student (Chandigarh University, 2026) with hands-on experience in mechanical design, prototyping, and system development. Skilled in SolidWorks, AutoCAD, and Python, with academic projects related to energy systems, automation, and applied mechanical engineering. Motivated to pursue graduate study to strengthen core mechanical engineering fundamentals through advanced coursework and supervised research.

EDUCATION

- **Bachelor of Engineering (B.E.) in Mechanical Engineering**
  Chandigarh University, Mohali, India
  Sep 2022 – Jun 2026 (Expected) *
  CGPA: 6.9 / 10 (up to 7th semester)

- **Higher Secondary School (Class XII)**
  Kendriya Vidyalaya, Mohali, India
  Apr 2021 – Mar 2022

CERTIFICATIONS & ONLINE COURSES

Design & Manufacturing

- Computer Aided Design — Indian Institutes of Technology (IIT) Guwahati
- SOLIDWORKS xDesign Specialization — Dassault Systèmes
- 3D Printing Applications — University of Illinois Urbana-Champaign

Energy / Thermal / Fluids

- Electric Vehicles and Mobility — École Nationale des Ponts et Chaussées
- Renewable Energy Technology Fundamentals — University of Colorado Boulder
- Computational Fluid Mechanics: Airflow Around a Spoiler

Mechanical Systems

- Machine Design Part I — Georgia Institute of Technology
- FEM: Linear & Nonlinear Analysis

PROJECTS

Dual-Axis Solar Tracking Energy System

- Designed and fabricated a solar-powered hybrid energy system integrating photovoltaic panels, optical reflectors, and thermoelectric modules.
- Implemented Arduino-based dual-axis tracking using sensor feedback to improve energy capture efficiency.
- Patent application under process.

Voice-Controlled Air Purifier

- Developed a smart air purification system incorporating HEPA filtration, UV-C sterilization, and real-time temperature monitoring.
- Integrated voice-based control and microcontroller logic for automated operation.

Portable Automated Ventilator

- Conceptualized a low-cost automated ventilator using AMBU-bag actuation with sensor-based control logic.
- Focused on emergency healthcare applications and basic mechanical-electronic integration.

Arduino-Based Smart Robotic Car

- Designed and built a robotic platform using Arduino, Bluetooth communication, and ultrasonic sensors.
- Implemented obstacle detection and basic autonomous navigation algorithms.

SKILLS

- **Core Mechanical Engineering:** Thermodynamics, Fluid Mechanics, Heat Transfer, Strength of Materials, Machine Design
- **Design & Engineering Tools:** SolidWorks, AutoCAD, Creo
- **Programming & Computational Skills:** Python, C++, MySQL (basic)
- **Software Tools:** MS Excel, Word, PowerPoint
- **Professional Skills:** Teamwork and collaboration, Technical communication, Problem-solving

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INDUSTRIAL TRAINING / INTERNSHIP

6-Week Industrial Training – SolidWorks

- Designed and modeled a drone system using advanced SolidWorks features.
- Gained hands-on experience in part design, assembly modeling, and basic simulation.

ACHIEVEMENTS & ACTIVITIES

- Student Member, American Society of Mechanical Engineers (ASME)
- Participated in workshop on Additive Manufacturing in Healthcare, CSIR-CSIO, Chandigarh

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INTERESTS

- Applied mechanical engineering and system-level problem solving
- Hands-on engineering projects and prototyping
- Programming for engineering applications (C, Python)